

FALL RIVER HARBOR, MA, USA

WMO: 997279

Lat: 41.7043N

Lon: 71.1640W

Elev: 10

StdP: 101.20

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -10.7	(c) -8.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 7.4	(c) 30.9	(d) N/A	(e) 29.4	(f) N/A	(g) 28.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.0	-0.5	0.5	4.0	10.5	15.5	20.3	24.6	23.2	19.7	13.7	8.0	3.9		
	DBStd	9.28	4.96	4.33	4.39	3.39	3.66	3.35	2.34	2.29	3.22	3.61	4.30	4.67		
	HDD10.0	1113	324	267	192	33	2	0	0	0	0	10	89	196		
	HDD18.3	2814	582	501	444	236	104	18	1	1	22	149	311	447		
	CDD10.0	1845	1	0	6	49	172	308	452	409	291	123	28	7		
	CDD18.3	506	0	0	0	1	16	77	194	151	63	4	0	0		
	CDH23.3	2808	0	0	0	13	75	373	1300	810	227	10	0	0		
	CDH26.7	639	0	0	0	2	12	79	357	154	34	1	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	1194	96	89	120	106	89	93	87	96	101	105	101	113		
	PrecMax	1458	161	181	373	203	199	286	230	170	234	325	165	204		
	PrecMin	916	43	45	4	37	31	3	29	21	19	8	19	34		
	PrecStd	153	31	36	74	44	36	73	40	42	51	66	39	43		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	12.7	18.9	25.4	28.4	31.5	33.4	31.9	30.0	24.7	19.7	16.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	12.8	10.3	14.5	19.8	25.4	29.0	31.4	29.6	27.3	22.2	17.5	14.6		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	9.7	8.5	12.2	17.5	23.4	27.1	30.0	28.4	25.7	20.4	15.9	13.1		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	6.9	6.6	10.2	15.8	21.3	25.5	28.7	27.2	24.4	19.0	14.6	11.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		6.9	7.0	7.3	8.3	8.3	7.8	7.4	7.2	7.7	6.8	6.9	6.4		
	5% DB	MCDBR	10.3	9.1	9.8	10.5	10.9	9.6	8.5	8.2	7.9	7.9	7.8	8.2		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.301	0.310	0.334	0.372	0.412	0.460	0.484	0.439	0.374	0.341	0.313	0.301		
	taud		2.506	2.487	2.466	2.388	2.311	2.215	2.156	2.271	2.429	2.530	2.566	2.561		
	Ebn at Noon		856	904	916	898	866	822	796	823	861	856	837	828		
	Edh at Noon		73	86	99	115	127	141	148	128	102	82	69	64		
(44) All-Sky Solar Radiation	RadAvg		1.84	2.72	3.73	4.80	5.33	5.80	5.99	5.38	4.38	2.96	2.02	1.54		
	RadStd		0.14	0.26	0.27	0.51	0.53	0.56	0.33	0.28	0.32	0.26	0.14	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+95	+47

Nomenclature: See separate page